Lab Session 02

Examine effects of sampling and aliasing on analog signals

Analog to digital conversion:

Most signals of practical interest, such as speech, biological signals, seismic signals, radar signals, sonar signals, and various communications signals such as audio and video signals, are analog. To process analog signals by digital means, it is first necessary to convert them into digital form, that is, to convert them to a sequence of numbers having finite precision. This procedure is called analog-to-digital (A/D) conversion, and the corresponding devices are called A/D converters (ADCs)

Conceptually, we view A/D conversion as a three step process. This process is illustrated in figure 2.1

Sampling:

This is the conversion of a continuous time signal to a discrete time signal obtained by taking "samples" of continuous time signal at discrete time instants. Thus, if $x_a(t)$ is the input to the sampler, the output is $x_a(nT) \equiv x(n)$, where T is called the sampling interval.

Quantization:

This is the conversion of discrete time continuous valued signal into a discrete time discrete valued (digital) signal. The value of each signal sample is represented by a value selected from a finite set of possible values. The difference between the un-quantized sample **x(n)** and the quantized output $x_q(n)$ is called the quantized error.

Coding:

In the coding process, each discrete value $x_q(n)$ is represented by a b-bit binary sequence.

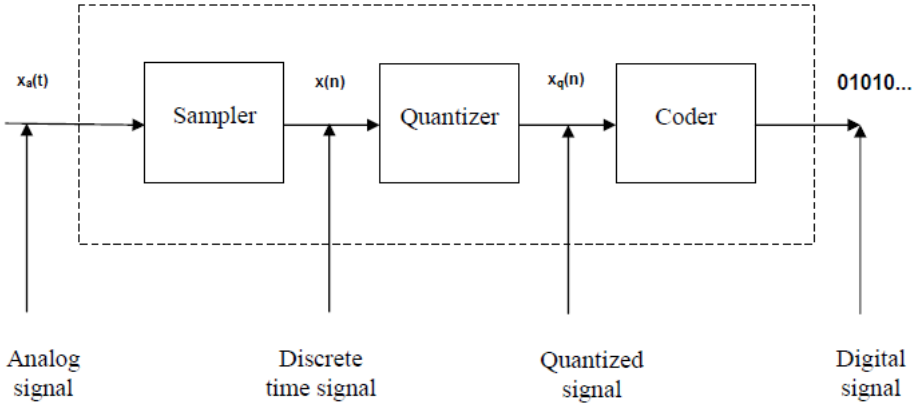


Figure 2.1: Basic parts of an analog-to-digital (A/D) converter.

There are many ways to sample an analog signal. We limit our discussion to periodic or uniform sampling, which is the type of sampling used most often in practice. This is described by the relation

$$x(n) = x_a(n) \quad -\infty < n < \infty \qquad (2.1)$$

Where x(n) is the discrete time signal obtained by "taking samples" of the analog signal $x_a(t)$ every T seconds. This procedure is illustrated in figure 2.2. The time interval T between successive samples is called the sampling period or sample interval and its reciprocal $1/T=F_s$ is

called the sampling rate (samples per second) or the sampling frequency (hertz).

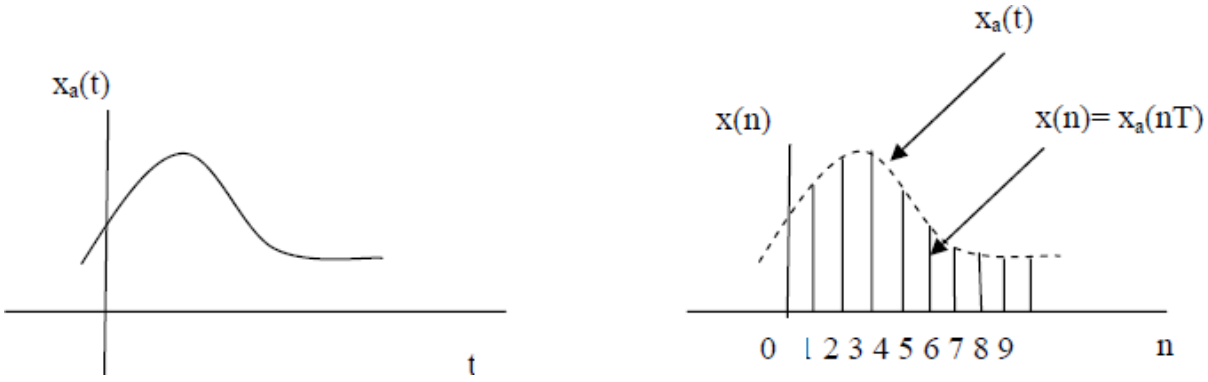


Figure 2.2: Periodic sampling of an analog signal

Periodic sampling establishes a relationship between the time variables t and n of continuous time and discrete time signals respectively. Indeed, these variables are linearly related through the sampling period T or, equivalently, through the sampling rate $F_s = 1/T$, as

$$t = nT = \frac{n}{F_s} \quad (2.2)$$

There exists a relationship between the variables F (or **Ω**) for analog signals and the frequency variable f (or ω) for discrete time signals. To establish this relationship, consider an analog sinusoidal signal of the form

$$x_a(t) = A\cos(2\pi Ft + \theta) \quad (2.3)$$

Which, when sampled periodically at a rate Fs =1/T samples per second, yields

$$x_a(nT) = x(n) = A\cos(2\pi FnT + \theta) = A\cos\left(\cos\left(\frac{2\pi NF}{F_s} + \theta\right)\right) \quad (2.4)$$

The frequency variables F and f are linearly related as

$$f = \frac{F}{F_s} \quad (2.5)$$

Or, equivalently, as

$$\omega = \Omega T \quad (2.6)$$

The relation in (1) justifies the name relative or normalized frequency, which is sometimes used to describe the frequency variable f. As (2.1) implies, we can use f to determine the frequency F in hertz only if the sampling frequency F_s is known.
The range of frequency variable F or Ω for continuous time sinusoids is

$$-\infty < F < \infty \quad (2.7)$$

$$-\infty < \mathbf{\Omega} < \infty \quad (2.8)$$

However, the situation is different for discrete time sinusoids.

$$\frac{-1}{2} < f < \frac{1}{2} \quad (2.9)$$

$$-\pi < \omega < \pi \qquad (2.10)$$

By substituting the given equations, we find that the frequency of the continuous time sinusoid when sampled at a rate Fs=1/T must fall in the range:

$$-\frac{1}{2T} = -\frac{Fs}{2} \leq F \leq \frac{Fs}{2} = \frac{1}{2T} \qquad (2.11)$$

From these relations we observe that the fundamental difference between the continuous time and discrete time signals is in their range of values of the frequency variables F and f, or Ω and ω. Periodic sampling of a continuous time signal implies a mapping of the infinite frequency range for the variable F (or Ω) into a finite frequency range for the variable f (or ω). Since the highest frequency in a discrete time signal is ω= π or f = 1/2, it follows that, with a sampling rate F_s, the corresponding highest value of F and Ω are

$$F_{max} = \frac{F_s}{2} = \frac{1}{2T} \qquad (2.12)$$

$$\Omega_{max} = \pi F_s = \frac{\pi}{T} \qquad (2.13)$$

Aliasing effect:

We only sample the signal at intervals. We don't know what happened between the samples. A crude example is to consider a 'glitch' that happened to fall between adjacent samples. Since we don't measure it, we have no way of knowing the glitch was there at all as shown in fig 9.1.

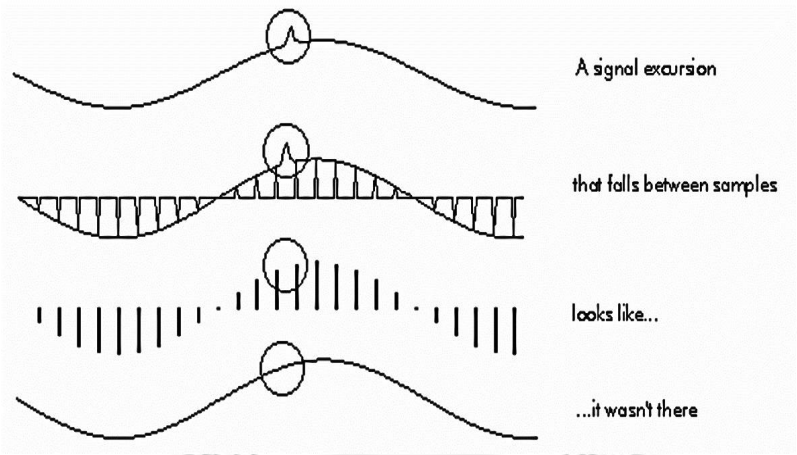


***Figure 2.3**: Basic view of sampling theorem*

In a less obvious case, we might have signal components that are varying rapidly in between samples. Again, we could not track these rapid inter-sample variations. We must sample fast enough to see the most rapid changes in the signal. Sometimes we may have some a priori knowledge of the signal, or be able to make some assumptions about how the signal behaves in between samples. If we do not sample fast enough, we cannot track completely the most rapid changes in the signal. Some higher frequencies can be incorrectly interpreted as lower frequencies

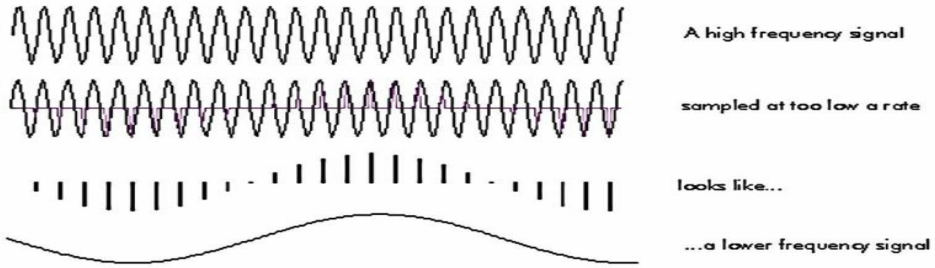


***Figure 2.4**: Overlapping of High frequency signal into Low frequency signal ($f_s \leq 2\ f_m$)*

In the fig 2.4:, the high frequency signal is sampled just under twice every cycle. The result is that each sample is taken at a slightly later part of the cycle. If we draw a smooth connecting line between the samples, the resulting curve looks like a lower frequency. This is called 'aliasing' because one frequency looks like another.

Note that the problem of aliasing is that we cannot tell which frequency we have - a high frequency looks like a low one so we cannot tell the two apart. But sometimes we may have some a priori knowledge of the signal, or be able to make some assumptions about how the signal behaves in between samples, that will allow us to tell unambiguously what we have.

Nyquist showed that to distinguish unambiguously between the signal frequencies we must sample faster than twice the frequency of the highest frequency component.

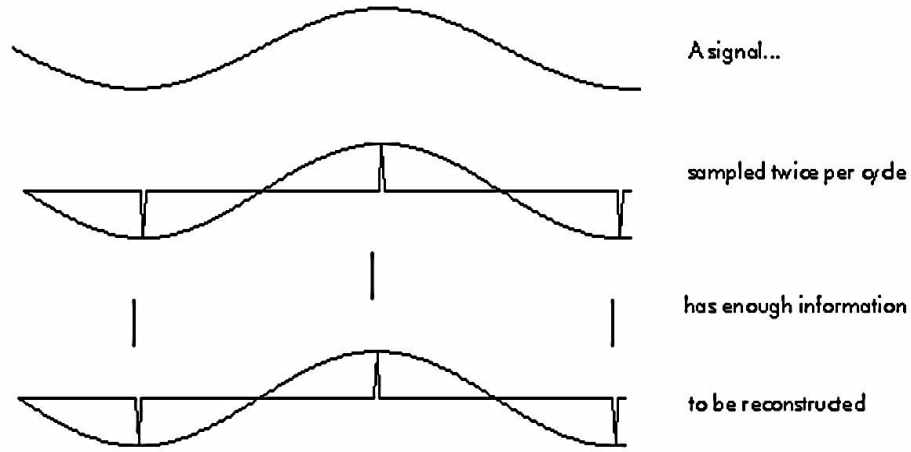


***Figure 2.5**: Reconstruction of signal with ($f_s = 2f_m$)*

In the diagram, the high frequency signal is sampled twice every cycle. If we draw a smooth connecting line between the samples, the resulting curve looks like the original signal. But if the samples happened to fall at the zero crossings, we would see no signal at all - this is why the sampling theorem demands we sample faster than twice the highest signal frequency as it will avoids aliasing.

The highest signal frequency allowed for a given sample rate is called the Nyquist frequency. Actually, Nyquist says that we have to sample faster than the signal bandwidth, not the highest frequency. But this leads us into multirate signal processing which is a more advanced subject. Nyquist showed that to distinguish unambiguously between all signal frequencies we must sample at least twice the frequency of the highest frequency component.

EXERCISE:

Task # 1: Write a script to generate a continuous time sinusoid signal with amplitude=2 for a time 0 to 1 sec and taken the frequency as a user input by using input command.

Task # 2: Write a script to convert continuous time signal of task#1 to a discrete time signal and take the sampling frequency as input which should be less than twice the maximum signal frequency in Hertz.

Task # 3: Write a script to convert continuous time signal of task#1 to a discrete time signal and take the sampling frequency as input which should be equal to twice the maximum signal frequency in Hertz.

Task # 4: Write a script to convert continuous time signal of task#1 to a discrete time signal and take the sampling frequency as input which should be greater than twice the maximum signal frequency in Hertz.

Task # 5: Write a script using subplot command to generate task #1, task#2, task#3 and task#4 plots. Analyze the four plots, what happen by increasing the sampling frequency? Write down your comments

Task # 6: Consider the analog signal

$$x_a(t) = 3\cos(100\pi t)$$

- Determine the minimum sampling rate required to avoid aliasing.
- Suppose that the signal is sampled at the rate Fs=200 Hz. What is the discrete time signal obtained after sampling? Plot the discrete-time signal.
- Suppose that the signal is sampled at the rate Fs=75 Hz. What is the discrete time signal obtained after sampling? Plot the discrete-time signal.


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Lab Session 03

Examine effects of quantization on discrete time continuous signals

Quantization:

Digital signal is a sequence of numbers (samples) in which each number denotes a finite number having finite precision.

The process of converting a discrete time continuous valued (DTCV) signal into a discrete time discrete valued (DTDV) signal is known as Quantization. In this process each sample expresses a value as finite (rather than infinite) number. The error introduced in representing the continuous valued signal by a finite set of discrete value levels is called Quantization error or Quantization noise.

We express the quantizer operation on the sampled signal x(n) as Q[x(n)] and the quantized signal can be expressed as xq(n) which denotes the sequence of quantized samples at the quantizer output. Hence

$$x_q = Q[x(n)] \qquad (3.1)$$

The quantization error is a sequence eq(n) defined as the difference between the quantized value and the actual sample value. Thus

$$e_q(n) = x_q(n) - x(n) \qquad (3.2)$$

It is obvious that the signal cannot be processed by using calculator or digital computer since only the first few samples can be stored and manipulated. For example, most calculators process number with only eight significant digits.

We assume that we want to use only one significant digit. To eliminate the excess digit we either simply discard them (truncation) or discard them by rounding the resulting number (rounding). The resulting quantized signals $x_q(n)$ are shown by the numerical illustrations in the Table 3.1 as below:

n	x(n) Discrete Time Signal	xq(n) Truncation	xq(n) Rounding	eq(n) = xq(n)-x(n)
0	1	1.0	1.0	0.0
1	0.9	0.9	0.9	0.0
2	0.81	0.8	0.8	-0.01
3	0.729	0.7	0.7	-0.029
4	0.6561	0.6	0.7	0.0439
5	0.59049	0.5	0.6	0.00951
6	0.531441	0.5	0.5	-0.031441
7	0.4782969	0.4	0.5	0.0217031
8	0.43046721	0.4	0.4	-0.03046721
9	0.387420489	0.3	0.4	0.012579511

Table 3.1: Error Computation and Analysis

The values allowed in the digital signal are called the quantization levels whereas the